

澳門大學
UNIVERSIDADE DE MACAU
UNIVERSITY OF MACAU

CISC 3024 Pattern Recognition – AI Assignment #2 Report

Bayesian Image Classification with IVON on Frozen StarNet Features Accuracy and Calibration on CIFAR-10

Ren Xianglin U-C3-2745-1

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Abstract

This report evaluates Improved Variational Online Newton (IVON), a Bayesian variational learning method presented at ICML 2024, for CIFAR-10 image classification. A previously trained StarNet-S050 is reused as a fixed feature extractor. A deterministic Adam-trained classifier is compared with temperature scaling, an IVON posterior-mean classifier, and Bayesian model averaging (BMA) over 32 posterior draws. All new classifiers use the same feature representation, data split, regularization coefficient, and paired initialization seeds. Learning rates and checkpoints are selected using validation negative log-likelihood (NLL), before test evaluation. Across three classifier seeds, Adam obtains 85.48% accuracy, NLL 0.4656, and expected calibration error (ECE) 1.01%; IVON BMA obtains 85.42%, 0.4730, and 2.59%, respectively. Averaging improves NLL slightly relative to IVON's posterior mean, but does not outperform Adam under this protocol. Independent checkpoint reloading reproduces the IVON BMA probabilities exactly in the recorded environment. The study documents algorithm selection, implementation, numerical checks, experimental limitations, and learning reflections. It is a last-layer adaptation with a reused backbone, rather than a reproduction of the paper's full-network experiments.

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1 Introduction and research question

Image classification requires assigning an image to a category, but many uses also require a meaningful estimate of confidence. Two classifiers can predict the same labels while allocating very different probabilities to those labels. An incorrect prediction with high confidence is penalized strongly by NLL, even when its effect on accuracy is only one misclassified image.

Assignment 1 studied a CIFAR-10 adaptation of StarNet, whose defining block uses element-wise multiplication between two feature branches [3]. This report retains that learned representation and investigates the probabilistic classifier placed on top of it. IVON is selected because it provides an explicit approximate weight posterior through variational learning [1]. The hypothesis is that posterior averaging may improve probability quality while retaining competitive classification accuracy. The experiment is designed to permit a negative answer.

The research question is: *under a fixed StarNet feature representation, does an IVON-trained Bayesian classifier improve test NLL and calibration over a deterministic classifier?* A second question asks whether averaging posterior predictions improves on using only IVON's mean weights. Keeping the feature extractor fixed isolates the head-level comparison and makes repeated runs practical. It also deliberately restricts the scope: uncertainty in the feature extractor is not modeled.

2 How AI selected the algorithm

The AI agent first read the assignment PDF and inspected the previous project, including its LaTeX source, model implementation, split indices, checkpoint, and experiment records. It then searched for recent Bayesian image-classification methods. Search queries included "2024 2025 Bayesian image classification last layer Laplace approximation paper" and "Variational Learning is Effective for Large Deep Networks IVON 2024 github official." The search also identified Laplace Redux (2021) and Variational Linearized Laplace Approximation (2024) [6, 7].

Selection used three practical criteria: an explicit probability or Bayesian foundation, a recent peer-reviewed publication, and an inspectable implementation that could support an experiment on the available hardware. IVON satisfied these criteria and enabled a direct posterior-mean versus posterior-average comparison. The publication year is 2024; it is a recent candidate relative to this October 2026 assignment, not a claim that it is the newest available method. The search was targeted, not a systematic review of all Bayesian classifiers.

The AI verified the publication through the ICML/PMLR entry and inspected the authors' official optimizer repository [2]. The original optimizer was downloaded with its GPL-3.0 license and SHA-256 hashes recorded. It was used without rewriting its update rules. The new experiment driver, validation, analysis, and report-generation code were authored by AI. This distinction separates integration work from inventing a new optimization algorithm.

3 Bayesian formulation and algorithm

3.1 A probabilistic linear classifier

Let $\phi(x) \in \mathbb{R}^{128}$ denote a fixed StarNet feature vector, standardized using only training-set statistics. Appending a constant permits the weights and biases of a ten-class linear head to be collected into $w \in \mathbb{R}^{1290}$. For logits $z_k(x, w)$, the categorical likelihood is

$$p(y = k \mid x, w) = \frac{\exp z_k(x, w)}{\sum_{j=1}^{10} \exp z_j(x, w)}. \tag{1}$$

The training data D update a weight prior according to Bayes' rule:

$$p(w \mid D) = \frac{p(D \mid w)p(w)}{p(D)}. \tag{2}$$

The softmax likelihood makes exact integration of this posterior inconvenient. We approximate it with a diagonal Gaussian $q(w) = \mathcal{N}(m, \mathrm{diag}(\sigma^2))$. The frozen backbone itself has no posterior in this experiment.

3.2 Variational objective and IVON state

With $N = 45{,}000$ training images, consider

$$\mathcal{J}(q) = \mathbb{E}_q \left[\frac{1}{N} \sum_{i=1}^{N} -\log p(y_i \mid x_i, w) \right] + \frac{1}{N} \mathrm{KL}(q \| p). \tag{3}$$

For the Gaussian prior $p(w) = \mathcal{N}(0, (N\delta)^{-1} I)$, its contribution includes $\delta \mathbb{E}_q[\|w\|^2]/2$. We set $\delta = 0.001$, corresponding to prior precision $N\delta = 45$. The deterministic comparator minimizes average cross-entropy plus the same quadratic penalty using Adam with coupled weight decay. The numerical penalty is matched; the two objectives still differ because IVON optimizes a distribution and includes its entropy.

The implementation maintains a mean, gradient momentum, and diagonal curvature state h. The sampling variance is

$$\sigma_j^2 = \frac{1}{N(h_j + \delta)}. \tag{4}$$

For one sampled weight vector and its minibatch gradient g, the implementation's curvature estimate can be written

$$\widehat{h} = (w - m) \odot g \oslash \sigma^2. \tag{5}$$

Here $\odot$ and $\oslash$ are element-wise operations. With curvature decay β_2, the positive-correction update is

$$h_{t+1} = \beta_2 h_t + (1 - \beta_2)\widehat{h}_t + \frac{(1 - \beta_2)^2}{2} \frac{(h_t - \widehat{h}_t)^2}{h_t + \delta}. \tag{6}$$

The mean step divides a bias-corrected momentum plus δm by $h_{t+1} + \delta$, with component-wise clipping at radius 1. The official implementation also rescales its supplied learning rate by $h_0 + \delta = 0.101$. Thus its nominal learning rate is not directly comparable to Adam's.

3.3 Prediction and uncertainty

Posterior-mean prediction evaluates $p(y \mid x, m)$. BMA instead estimates

$$\bar{p}_k(x) = \frac{1}{S} \sum_{s=1}^{S} p(y = k \mid x, w^{(s)}), \qquad w^{(s)} \sim q(w). \tag{7}$$

Softmax probabilities are averaged, rather than logits or hard class labels. The final prediction is $\arg\max_k \bar{p}_k(x)$. In general, $\mathbb{E}_q[\mathrm{softmax}(z)] \neq \mathrm{softmax}(\mathbb{E}_q[z])$. The comparison between BMA and posterior-mean prediction measures the practical effect of this distinction for the same fitted posterior.

A distribution over weights can represent parameter uncertainty, but neither posterior averaging nor a Bayesian interpretation guarantees calibration on an arbitrary dataset. Predictive entropy also combines several sources of uncertainty; we do not interpret it as a pure measure of epistemic uncertainty or claim out-of-distribution detection without a corresponding experiment.

4 How AI implemented and checked the experiment

The AI copied the previous StarNet-S050 implementation, checkpoint, split indices, and license into a separate assignment-2 project. The original project remained unchanged. The backbone was put in evaluation mode so that batch normalization statistics were fixed. Each image was normalized using the original CIFAR-10 constants, and the 128-dimensional pooled features were cached. Feature centering and scaling were fitted only on the 45,000 training vectors.

Each new classifier has 1,290 trainable parameters. Paired seeds initialize the two methods with the same zero biases and normally distributed weights with standard deviation 0.01. A separate random-number generator controls minibatch order. Validation posterior sampling saves and restores the training RNG state, so evaluation does not change subsequent training samples. Model state and IVON optimizer state are saved together to preserve both posterior means and variances.

The implementation workflow was:

1. inspect the assignment, previous project, publication, and official code;
2. save a protocol specifying data use, candidate learning rates, seeds, metrics, checkpoint selection, and posterior sample counts;
3. verify normalization, posterior restoration, finite updates, positive precision, and metric reference cases;
4. extract features and run a two-epoch smoke test on 2,048 training vectors;
5. run the declared validation search and three-seed formal experiment;
6. freeze chosen learning rates before computing the new test metrics;
7. save probabilities, reload checkpoints independently, and regenerate tables and figures from measured files.

For a uniform ten-class prediction, the metric test recovers NLL $\log 10$, Brier score 0.9, and ECE zero on ten balanced labels. Perfect one-hot predictions recover zero NLL and Brier score. The posterior sampling check confirms exact restoration of mean parameters. These tests examine mathematical properties instead of merely checking that a script runs.

The smoke test completed for both methods. A later audit started a separate Python process, reloaded the saved IVON checkpoints, and reproduced all three saved BMA probability arrays with maximum absolute difference 0. This verifies the saved posterior and prediction path on the recorded software/hardware; it does not guarantee bitwise equality on every other device.

5 Experimental settings

5.1 Data and provenance

CIFAR-10 contains 60,000 color images of size 32×32 across ten classes [4]. The official training set is split into 45,000 training and 5,000 validation images using the saved class-stratified seed-42 indices from Assignment 1. The official 10,000-image test set is evaluated after new model selection. Training and validation indices are checked for disjointness and complete coverage of the 50,000-image training set.

The reused backbone was originally trained for 50 epochs with image augmentation, and its checkpoint was selected by validation accuracy. The current head training uses fixed, unaugmented features. The old validation set therefore participates in both historical backbone selection and current head selection. The test set was also already evaluated in Assignment 1. This

is a follow-up study on a known benchmark, not a claim of a newly untouched test set. No new head setting was chosen by its test performance.

5.2 Matched comparisons and selection

Table 1: Predeclared head-training and evaluation settings.

Setting	Value
Feature extractor	Frozen StarNet-S050, 414,010 total parameters
New classifier	Linear $128 \to 10$; 1,290 parameters
Head seeds	42, 43, 44; one shared backbone and data split
Epoch budget / batch size	60 / 1,024; final partial batch retained
Adam candidates	Learning rates 0.001 and 0.01; defaults $\beta = (0.9, 0.999)$
IVON candidates	Nominal learning rates 0.05 and 0.2
IVON state	$h_0 = 0.1$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, clip radius 1
Regularization	Coupled coefficient $\delta = 0.001$ for weights and biases
IVON effective sample size	45,000; one weight draw per training minibatch
Schedule	Cosine decay to 1% of initial learning rate
Checkpoint selection	Lowest validation NLL at epochs 1, 5, 10, …, 60
Posterior draws	20 for validation; 32 for main test evaluation
Temperature scaling	Scalar $T \in [0.05, 20]$ fitted on validation NLL

Both candidate learning rates are evaluated at seed 42. The selected rate is then used for seeds 43 and 44, with each run selecting its checkpoint by validation NLL. IVON selection uses 20-draw BMA. The winning rates are 0.001 for Adam and 0.05 for IVON. All six selected runs choose epoch 5; the entire 60-epoch budget was nevertheless executed. The higher-rate candidates select epoch 1 and have worse validation NLL. Sampling uses fixed evaluation seeds. Validation is reused for temperature fitting, so its apparent calibration improvement should not be interpreted as an independent estimate of generalization.

Temperature scaling replaces logits z by z/T, with a positive scalar T [5]. It preserves each image's predicted class, while modifying its probability vector. The fitted temperatures for seeds 42, 43, and 44 are 0.9656, 0.9640, 0.9679, respectively.

5.3 Metrics, uncertainty estimates, and timing

We report accuracy, mean NLL, multiclass Brier score, and 15-bin ECE:

$$\mathrm{NLL} = -\frac{1}{n}\sum_{i} \log p_{i,y_i}, \tag{8}$$

$$\mathrm{Brier} = \frac{1}{n}\sum_{i}\sum_{k=1}^{10}(p_{ik} - \mathbf{1}[y_i = k])^2, \tag{9}$$

$$\mathrm{ECE} = \sum_{b=1}^{15}\frac{|B_b|}{n}|\mathrm{accuracy}(B_b) - \mathrm{confidence}(B_b)|. \tag{10}$$

The Brier score is summed over classes, not divided by ten. Confidence is the largest predicted probability. ECE bins have equal width over $[0, 1]$, with empty bins omitted and confidence 1 assigned to the final bin. Probabilities are clipped below at 10^{-12} only when taking logarithms. ECE is descriptive and bin-dependent, so NLL and Brier provide additional evidence.

Reported standard deviations are sample standard deviations across three head seeds. For paired NLL differences, image-wise losses are first averaged across the three seeds, then 10,000

image indices are resampled with replacement 2,000 times using bootstrap seed 20261011. Percentile intervals quantify test-image variation conditional on these fitted models, not uncertainty across independently trained backbones or new datasets.

The machine uses an NVIDIA GeForce RTX 3070 Laptop GPU and PyTorch 2.9.1+cu128. Training uses FP32, deterministic PyTorch algorithms, no TF32, and fixed random seeds. Feature extraction takes 10.2 seconds. Head-fit times include the full 60 epochs and periodic validation, but exclude feature extraction. Head-only inference times are measured on all 10,000 cached test vectors, after five warm-ups, as the median of 20 synchronized repetitions per seed. The table reports the mean of those medians. This is a local implementation measurement, not an end-to-end image-processing benchmark.

6 Results

6.1 Accuracy and probability quality

Table 2: Test results: mean ± sample SD across three head seeds. Accuracy and ECE are percentages. Lower is better except for accuracy.

Method	Accuracy	NLL	Brier	ECE
Adam	85.48 ± 0.06	0.4656 ± 0.0005	0.2128 ± 0.0002	1.01 ± 0.01
Adam + TS	85.48 ± 0.06	0.4657 ± 0.0005	0.2127 ± 0.0002	1.03 ± 0.10
IVON mean	85.47 ± 0.02	0.4738 ± 0.0004	0.2136 ± 0.0001	2.60 ± 0.04
IVON BMA (32)	85.42 ± 0.01	0.4730 ± 0.0004	0.2134 ± 0.0001	2.59 ± 0.06

Adam has the lowest mean NLL and ECE in Table 2. IVON BMA is 0.06 percentage points lower in accuracy and about 0.00736 higher in NLL than Adam. Thus this experiment does not support an overall probability-quality advantage for IVON. Temperature scaling leaves accuracy unchanged, slightly improves mean Brier score, and slightly worsens test NLL and mean ECE. Optimization of validation NLL does not guarantee an improvement in every test metric.

Within IVON, BMA lowers mean NLL from 0.4738 to 0.4730 and slightly improves Brier and mean ECE relative to the posterior mean. Its accuracy is slightly lower. The averaging effect is real in these saved predictions but small compared with the gap to the deterministic comparator.

6.2 Calibration and selective prediction

The seed-42 reliability diagrams agree with the numerical ECE comparison: IVON has larger weighted deviations between confidence and empirical accuracy. These plots are not proof of calibration at every input or within every class. The displayed seed is fixed in advance by convention, rather than chosen for the most favorable picture.

At 80% coverage, Adam's error among retained images is 6.300%, compared with 6.3625% for IVON BMA. At 50% coverage, the values are 1.42% and 1.46%. The methods therefore provide similar confidence ranking in this example; IVON does not show a selective-prediction advantage. Temperature scaling can change the ordering of maximum confidence across images even though it preserves the predicted label of each image.

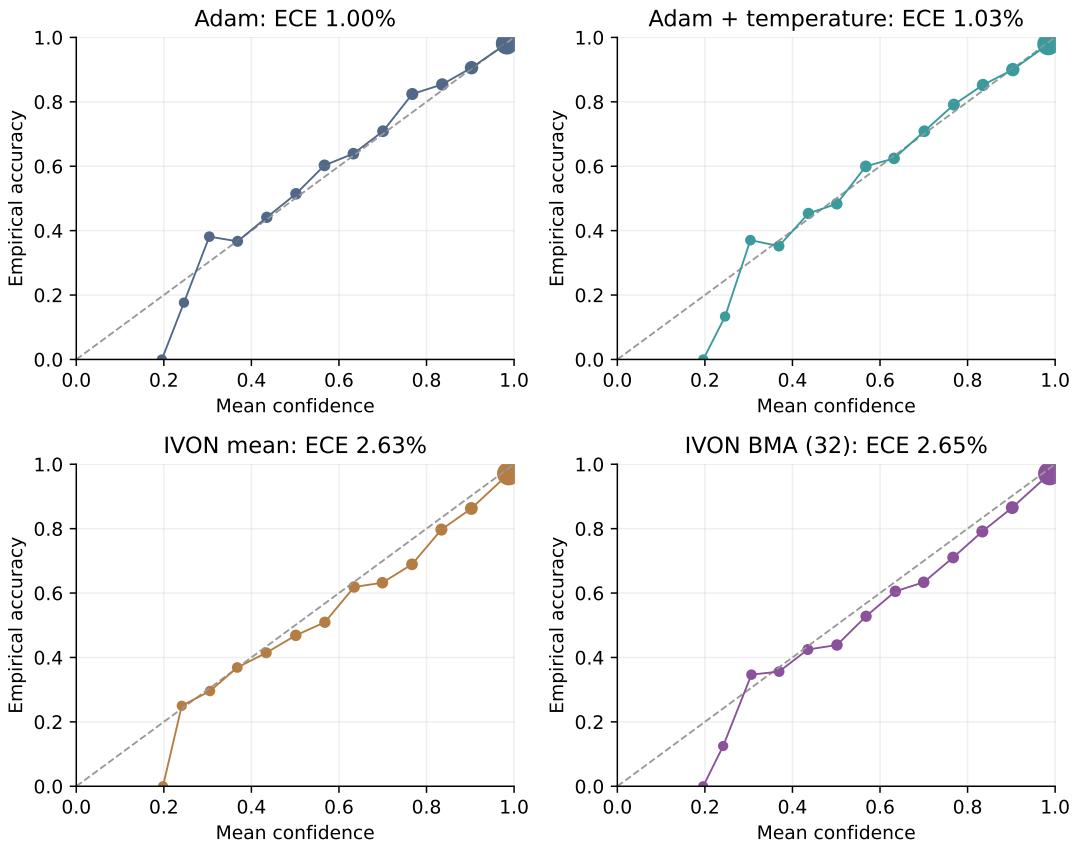


Figure 1: Seed-42 reliability diagrams. A point on the diagonal has matching confidence and empirical accuracy within its bin. Marker size reflects bin population. Sparse bins should not be overinterpreted.

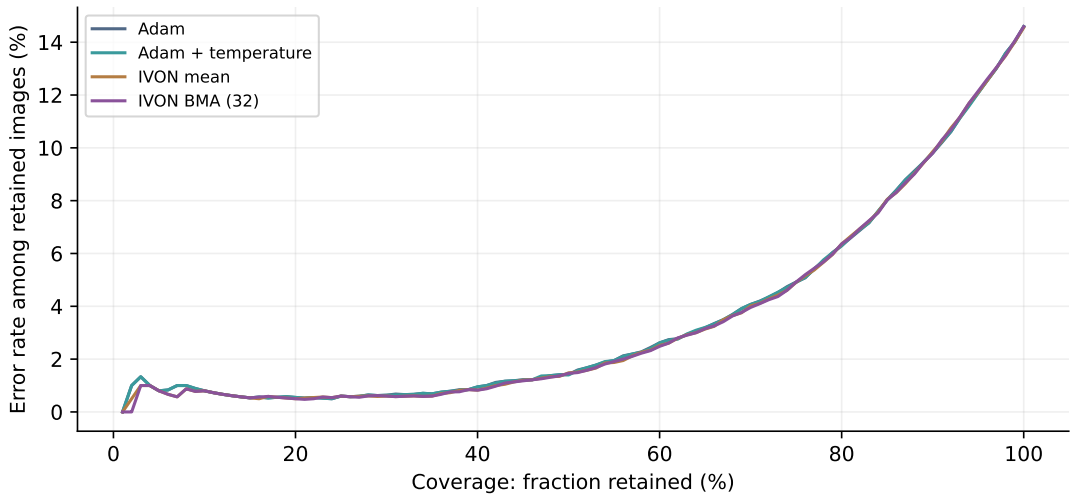


Figure 2: Seed-42 selective error versus coverage. Images are retained in descending maximum-probability order. This is a descriptive test-set curve, not a deployed abstention threshold selected on test data.

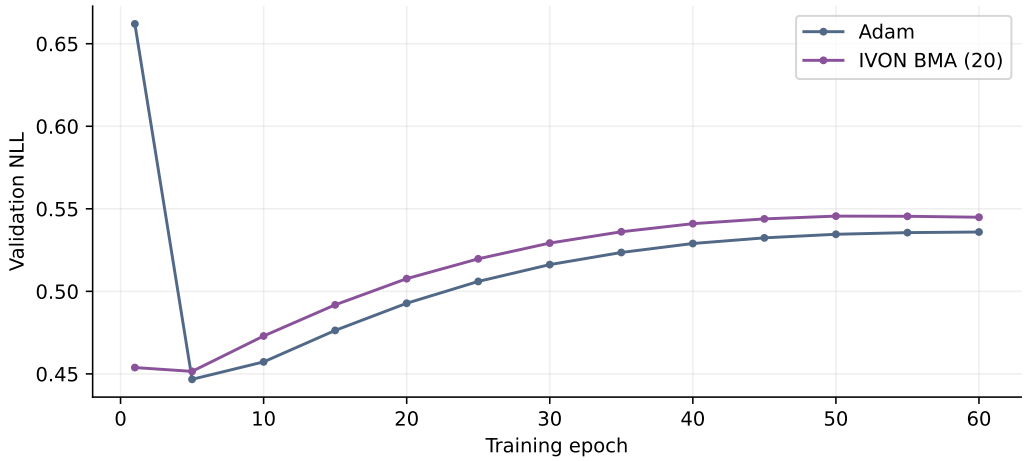


Figure 3: Seed-42 validation NLL at the selected learning rates. The best recorded checkpoint is epoch 5 for both methods; later epochs are not chosen.

6.3 Training trajectory and posterior sample count

The validation trajectories show why the final epoch is not used automatically. Training a head longer on a representation already fitted to the same training images can worsen held-out probability estimates. Checkpoint selection prevents the 60-epoch budget from forcing use of a poorer final model. The coarse five-epoch inspection interval may miss a better checkpoint between recorded epochs.

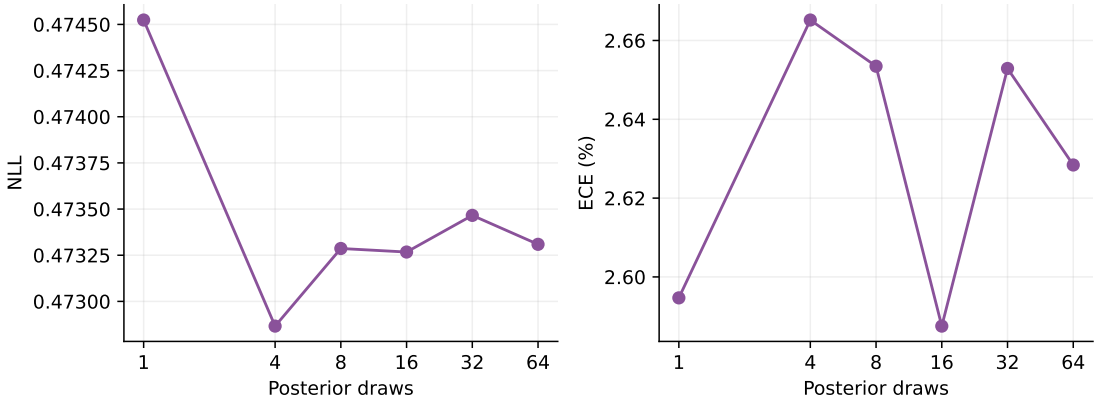


Figure 4: Seed-42 posterior sample sensitivity, using prefixes of one fixed sampling sequence. The main 32-draw setting was specified before this diagnostic.

The sample-count diagnostic evaluates 1, 4, 8, 16, 32, and 64 draws without retraining or changing the selected checkpoint. Monte Carlo estimates can fluctuate as samples are added, and ECE need not improve monotonically. This single sampling sequence does not establish an optimal draw count or quantify all Monte Carlo variability. No setting is changed in response to this test plot.

6.4 Paired uncertainty and computational cost

Table 3: Paired mean NLL differences and conditional 95% bootstrap intervals. Negative differences favor IVON BMA.

Comparison	Difference	95% interval
BMA minus Adam	+0.007362	[+0.003915, +0.010765]
BMA minus Adam + TS	+0.007259	[+0.004298, +0.010245]
BMA minus IVON mean	-0.000802	[-0.000986, -0.000611]

The intervals are consistent with a small NLL improvement from averaging the IVON posterior and a larger disadvantage relative to Adam. They should not be read as evidence that IVON is generally inferior: they condition on one learned feature extractor, one split, and this narrow training protocol.

Table 4: Measured compute cost. Fit includes 60 head-training epochs and validation. Inference covers all 10,000 cached test feature vectors.

Method	Head fit (s)	Head inference (ms)
Adam	1.923	0.095
Adam + TS	1.923	0.094
IVON mean	3.115	0.089
IVON BMA (32)	3.115	8.028

The temperature row shares the Adam fit; its scalar fitting time is not included. The two IVON rows share one trained posterior. BMA repeats the head calculation and includes sampling and RNG-management overhead. Since the head is small, these timings are especially sensitive to Python and GPU-launch overhead. Caching the backbone makes the experiment inexpensive, but the original backbone training and current feature extraction remain necessary costs.

7 Discussion and limitations

The main hypothesis is not supported under this setting. Bayesian inference provides a principled way to incorporate parameter uncertainty, but predictive quality depends on the approximate posterior, representation, prior, optimizer settings, and selection protocol. The study cannot isolate which of these causes the observed disadvantage. In particular, the limited two-rate search and fixed prior do not establish the best achievable result for either method.

Several scope restrictions are important:

- **Frozen representation.** Only the final 1,290 parameters are Bayesian; feature uncertainty and end-to-end adaptation are excluded. The comparison is not a reproduction of full-network IVON training in the original paper.
- **Shared historical selection.** The backbone and current heads use the same validation split at different stages. Repeated selection can adapt to that validation set. A nested split or independently trained backbone would provide a stronger future evaluation.
- **Three conditional repeats.** Head seeds vary initialization, data order, and IVON sampling. They do not vary the backbone, split, or dataset. Small SD therefore does not imply universal stability.

- **Approximation and budget.** A diagonal Gaussian cannot capture weight correlations or multiple modes. Hyperparameters other than learning rate are not searched. Early selected checkpoints may constrain posterior adaptation, but testing that explanation would require a separate ablation.
- **Metric scope.** ECE depends on binning, and accuracy is insensitive to most probability changes. No robustness, corruption, fairness, or out-of-distribution conclusion is supported by these experiments.
- **Known benchmark.** The old test results were already available. The current protocol avoids new test-based tuning, but does not turn this reused test set into an unseen external validation dataset.

A useful next study would predeclare a separate validation split and evaluate full-network IVON, several priors, and multiple independently trained backbones. That would test a broader question at higher computational cost. Retuning the current method after viewing these test results and reporting only an improved variant would weaken the evidential value of this comparison.

8 Learning reflection

This assignment connects Bayes' rule to a working image-classification system. I learned to distinguish a distribution over weights from the softmax probabilities produced by a single set of weights. A softmax layer already outputs probabilities, but that alone does not make its training Bayesian. In the present implementation, the Bayesian component is the approximate posterior and the predictive averaging over its samples.

I also learned that accuracy and probability quality answer different questions. The four methods have similar accuracies, while their NLL and ECE differ more clearly. Temperature scaling demonstrates the distinction particularly well: the labels remain unchanged, but the probabilities and Brier score change. A small improvement in one metric should not be described as an improvement in every aspect of uncertainty estimation.

Another lesson is that experimental controls are part of the algorithm evaluation. Matching the feature representation, initialization, regularization, and data order makes the comparison easier to interpret. Separating validation selection from test reporting reduces one source of optimistic conclusions, although reuse of the historical backbone and validation set still imposes limitations.

Finally, AI can write and execute a substantial part of the workflow, but its outputs need inspectable evidence. In this project the useful checks were saved probabilities, explicit metric definitions, numerical reference tests, independent checkpoint reloading, and source provenance. A negative experimental result is valuable when it is reproducible and its scope is stated accurately. These reflections are an AI-assisted draft for the student to review against their own understanding before submission.

9 Conclusion and source-code availability

The project implemented an AI-assisted last-layer Bayesian classifier using IVON and evaluated it on fixed StarNet features. Under the declared protocol, Adam achieved better NLL and ECE than IVON BMA, while their accuracies differed by only 0.06 percentage points. Averaging the IVON posterior provided a modest NLL benefit relative to its mean weights. The evidence supports this limited finding and does not establish a general ranking of Bayesian and deterministic learning methods.

The accompanying project contains the AI-authored experiment and analysis scripts, vendored IVON source with its license, the reused backbone, selected head checkpoints, split indices, prediction arrays, raw metrics, and this LaTeX source. The authors' reference implementation is available at https://github.com/team-approx-bayes/ivon. **The assignment-specific source-code webpage has not yet been published.** The local source archive is prepared, but the assignment's requirement for a webpage link remains outstanding until this project is uploaded and its URL is inserted here. The official repository URL is not a substitute for that project URL.

References

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A Per-seed results and reproducibility

Table 5: Individual test results; accuracy and ECE in percent.

Seed	Method	Accuracy	NLL	Brier	ECE
42	Adam	85.41	0.46618	0.21307	1.003
42	Adam + TS	85.41	0.46629	0.21289	1.028
42	IVON mean	85.45	0.47408	0.21365	2.631
42	IVON BMA (32)	85.41	0.47347	0.21357	2.653
43	Adam	85.51	0.46518	0.21268	1.007
43	Adam + TS	85.51	0.46526	0.21248	0.930
43	IVON mean	85.47	0.47334	0.21346	2.562
43	IVON BMA (32)	85.43	0.47262	0.21335	2.525
44	Adam	85.52	0.46554	0.21275	1.027
44	Adam + TS	85.52	0.46566	0.21258	1.122
44	IVON mean	85.49	0.47398	0.21356	2.621
44	IVON BMA (32)	85.42	0.47291	0.21337	2.581

The result tables and plots are generated from JSON and NPZ artifacts, rather than manually entered measurements. The implementation hashes the official IVON files and the reused backbone. The backbone SHA-256 is

```
040c77f7c885cc668d1c05563a2afddd
ed99a9916fc242e75eca69987f7ab364
```

The current batched extraction evaluates the original checkpoint at 85.19% test accuracy, whereas Assignment 1 recorded 85.18%. This one-image discrepancy is retained transparently: the current extraction uses vectorized GPU preprocessing and a different batch size from the earlier pipeline. Its precise numerical cause has not been isolated, so exact reproduction of the historical evaluation is not claimed. This reference is not a new training run and is not used to select a head. The independent audit above concerns the new IVON probabilities and reproduces those arrays exactly in the current pipeline.

The minimal reproduction sequence, after installing the dependencies and obtaining the CIFAR-10 Python dataset, is:

```
python scripts/experiment.py --stage smoke --data-root DATA_ROOT
python scripts/experiment.py --stage full --data-root DATA_ROOT
python scripts/experiment.py --stage audit --data-root DATA_ROOT
python scripts/retime.py
python scripts/analyze.py
python scripts/build_report.py
```

The shipped archive already contains the backbone and split indices. It excludes the CIFAR-10 image files, virtual environments, and temporary rendering cache. Feature caches may be regenerated. Exact reruns require the same data, package versions, GPU behavior, and evaluation seeds; numerical equality across hardware is not assumed. The scripts are designed for complete runs and evaluation, not for resuming interrupted IVON training midway through an epoch.

B AI workflow record and requirement mapping

Table 6: Evidence corresponding to the assignment requirements.

Requirement	Evidence in this project
How AI finds the algorithm	Section 2: queries, candidates, selection rationale, publication and source checks
Algorithm description	Section 3: likelihood, posterior, variational objective, IVON state, posterior predictive average
How AI implements it	Section 4: code integration, checks, smoke test and audit
Experimental settings/results	Sections 5–7, raw JSON/NPZ, generated figures
What was learned	Section 8: reflection draft for student review
Webpage link of source code	Local archive prepared; project webpage publication and URL insertion still required

The actual workflow encountered a local execution-sandbox initialization failure; approved shell execution was used to access the specified project and run the experiment. No experimental metric was invented to bypass that obstacle. The official IVON implementation downloaded successfully, GPU/data checks passed, and the numerical smoke test passed before formal training. A timing review identified unnecessary random-state bookkeeping on the deterministic inference path. This was removed and inference timing was repeated; model selection and saved probabilities were unchanged. The independent reload audit was retained.

The AI authored the new experiment, plotting, statistical analysis, and report scripts. It reused the previously generated StarNet adaptation and the authors' licensed IVON optimizer. The supplied project logs summarize actual actions; they are not presented as verbatim transcripts of unrecorded interactions.